

GitLoud Private Repository Integration

Simple implementation report - June 3, 2026

Goal

GitLoud originally worked mainly for public GitHub repositories. The goal was to support private PR and commit generation without asking users for personal access tokens and without using one global GitHub token for everyone.

Problem In The Old Flow

The old flow was:

```
web form -> POST /pr -> GitHub fetchers -> Octokit with GITHUB_TOKEN
```

- Public repositories worked.
- Private repositories failed with GitHub 404.
- The code had a direct block that rejected private repositories.
- There was no database table to store GitHub App installations.

Solution Used

We integrated a GitHub App installation model. A user installs the GitLoud GitHub App on selected repositories. The backend stores only the installation information, then creates a short-lived installation access token whenever it needs to read a private PR or commit.

Main Backend Changes

- Added GitHub App helper code to create installation tokens.
- Added routes for GitHub install URL, callback, installation list, sync, and delete.
- Stored GitHub installations in the database.
- Synced selected repositories into the database after callback/update.
- Changed PR/commit generation to use the installation token for matching private repos.
- Kept the global GitHub token only as a public repository fallback.
- Removed the old hard block that rejected private repositories.

Database Changes

- Added a user plan field so private repo support can be limited to Pro users.
- Added `GitHubInstallation` to store installation id and account details.
- Added `GitHubInstallationRepository` to store selected repository access.

Frontend Changes

- Added a Settings page.
- Added GitHub connection UI.
- Added Connect GitHub and Sync repositories actions.
- Added dropdown view for installed/selected repositories.
- Added delete buttons beside repositories to remove stored local access mapping.

Security Rules Followed

- No user personal access tokens are stored.
- No GitHub tokens are sent to the browser.
- Installation access tokens are short-lived and generated on the backend.
- The database stores installation ids and repo mapping, not temporary access tokens.
- Private repo support is allowed only for Pro users.

Final Result

Private repository PR/commit generation now works through the GitHub App. The user installs the app, GitLoud syncs repository access, and generation uses a backend-only installation token.